

s_lo1r: Model and Simulation Results

May 18, 2026

$$V^{ND}(b, l, g, s, \varepsilon) = \max_{b', l'} \{ u(c^{ND}) + \beta g^{1-\gamma} \mathbb{E}[\max\{V^{ND}(b', l', g', s', \varepsilon'), V^D(b', l', g', s', \varepsilon')\} \mid g, s] \}$$

$$c^{ND} = y(g, \varepsilon) - b - l + n(b', l', g, s) + n_l^{ND}(l', g)$$

$$V^D(b, l, g, s, \varepsilon) = \max_{l'} \left\{ u(c^D) + \beta g^{1-\gamma} \mathbb{E} \left[\theta V^D \left(\frac{b}{g}, l', g', s', \varepsilon' \right) + (1 - \theta) \max \left\{ V^{ND} \left(\kappa \frac{b}{g}, l', g', s', \varepsilon' \right), \right. \right. \right.$$

$$c^D = \phi_g y(g, \varepsilon) - l + n_l^D(l', g)$$

$$n_l^{ND}(l', g) = \frac{g l'}{R_l^{ND}}, \quad n_l^D(l', g) = \frac{g l'}{R_l^D}$$

$$d(b, l, g, s, \varepsilon) = \mathbf{1} \{ V^D(b, l, g, s, \varepsilon) > V^{ND}(b, l, g, s, \varepsilon) \}$$

$$Q(b, l, g, s, \varepsilon) = (1 - d) + d X(b, l, g, s, \varepsilon)$$

$$X(b, l, g, s, \varepsilon) = \frac{1}{1 + r^*} \mathbb{E}[\theta X' + (1 - \theta) ((1 - e')X' + e' \kappa Q') | g, s]$$

$$e'(b', l', g', s', \varepsilon') = \mathbf{1} \{ V^{ND}(\kappa b', l', g', s', \varepsilon') \geq V^D(b', l', g', s', \varepsilon') \}$$

$$R(b', l', g, s) = \frac{1 + r^*}{\mathbb{E}[Q' | g, s]}, \quad n(b', l', g, s) = \frac{g b'}{R(b', l', g, s)}$$

$$p^{def}(b', l', g, s) = \mathbb{E}[d' | g, s]$$

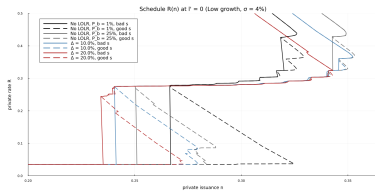
$$p^{def}(b', l', g, s) \leq \bar{p}_b, \quad n(b', l', g, s) \leq \bar{n}_g, \quad l' \leq \bar{l}^{ND}, \bar{l}^D$$

Simulation Table

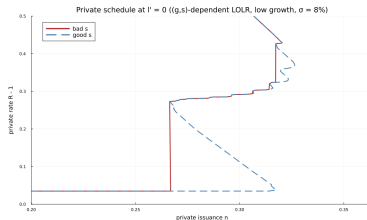
Table: Sunspot moments with and without LOLR access

Moment	No LOLR		$\Delta = 1\%$, $P_b = 25\%$		$\Delta = 5\%$, $P_b = 25\%$		$\Delta = 10\%$, $P_b = 25\%$		$\Delta = 20\%$, $P_b = 25\%$		Main
	$P_b = 1\%$	$P_b = 25\%$	$\sigma = 4\%$	$\sigma = 8\%$	$\sigma = 4\%$	$\sigma = 8\%$	$\sigma = 4\%$	$\sigma = 8\%$	$\sigma = 4\%$	$\sigma = 8\%$	$\sigma = 8\%$, $\Delta_{ND, LB} = 20\%$, $\Delta_{else} = 5\%$
<i>First moments</i>											
avg(spread)	0.0016	0.0916	0.0909	0.0921	0.0912	0.0920	0.0925	0.0903	0.0921	0.0922	0.0021
avg(qb/y)	0.3354	0.3652	0.3636	0.3664	0.3649	0.3663	0.3660	0.3634	0.3669	0.3667	0.3334
avg(f/y)	0.3511	0.4580	0.4567	0.4589	0.4576	0.4592	0.4589	0.4560	0.4601	0.4602	0.3492
avg(n/y)	0.3387	0.4044	0.4035	0.4051	0.4042	0.4054	0.4050	0.4031	0.4061	0.4062	0.3366
avg(n.l/y)	0.0000	0.0000	0.0084	0.0078	0.0437	0.0411	0.0931	0.0887	0.1865	0.1791	0.0473
avg(l/y)	0.0000	0.0000	0.0090	0.0086	0.0461	0.0452	0.0978	0.0970	0.1959	0.1953	0.0524
avg(b/y)	0.3477	0.4131	0.4110	0.4146	0.4126	0.4144	0.4143	0.4106	0.4151	0.4149	0.3457
avg(tb/y)	0.0091	0.0087	0.0080	0.0103	0.0107	0.0131	0.0140	0.0159	0.0184	0.0249	0.0143
default rate	0.0047	0.1505	0.1536	0.1473	0.1517	0.1500	0.1496	0.1526	0.1495	0.1516	0.0053
<i>Low-growth state</i>											
avg(spread)	0.0020	0.0006	0.0014	0.0008	0.0013	0.0007	0.0008	0.0006	0.0019	0.0015	0.0011
avg(qb/y)	0.3519	0.2998	0.2998	0.2995	0.3004	0.2997	0.2998	0.2996	0.2999	0.2994	0.3483
avg(f/y)	0.3509	0.2971	0.2981	0.2975	0.2985	0.2978	0.2979	0.2978	0.2997	0.2984	0.3450
avg(n/y)	0.3384	0.2869	0.2875	0.2871	0.2879	0.2875	0.2876	0.2875	0.2888	0.2878	0.3330
avg(n.l/y)	0.0000	0.0000	0.0089	0.0086	0.0445	0.0429	0.0939	0.0905	0.1876	0.1809	0.0622
avg(l/y)	0.0000	0.0000	0.0100	0.0100	0.0498	0.0498	0.1051	0.1051	0.2101	0.2102	0.0610
avg(b/y)	0.3649	0.3100	0.3106	0.3102	0.3113	0.3104	0.3105	0.3102	0.3109	0.3102	0.3609
avg(tb/y)	0.0265	0.0232	0.0242	0.0244	0.0286	0.0298	0.0342	0.0373	0.0446	0.0517	0.0267
default rate	0.0125	0.3902	0.3880	0.3874	0.3889	0.3915	0.3944	0.3852	0.3909	0.3963	0.0138
<i>High-growth state</i>											
avg(spread)	0.0013	0.1108	0.1106	0.1109	0.1108	0.1109	0.1112	0.1106	0.1109	0.1108	0.0028
avg(qb/y)	0.3255	0.3790	0.3777	0.3802	0.3790	0.3801	0.3795	0.3779	0.3809	0.3805	0.3242
avg(f/y)	0.3511	0.4917	0.4917	0.4922	0.4923	0.4926	0.4917	0.4918	0.4935	0.4935	0.3518
avg(n/y)	0.3388	0.4291	0.4291	0.4294	0.4295	0.4298	0.4289	0.4292	0.4306	0.4306	0.3388
avg(n.l/y)	0.0000	0.0000	0.0083	0.0076	0.0435	0.0408	0.0930	0.0883	0.1863	0.1787	0.0381
avg(l/y)	0.0000	0.0000	0.0087	0.0084	0.0453	0.0443	0.0963	0.0952	0.1929	0.1923	0.0472
avg(b/y)	0.3373	0.4347	0.4331	0.4361	0.4347	0.4360	0.4354	0.4333	0.4368	0.4364	0.3365
avg(tb/y)	-0.0015	0.0057	0.0044	0.0074	0.0068	0.0097	0.0099	0.0110	0.0129	0.0194	0.0067
default rate	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Selected Private Schedule: Baseline vs V2



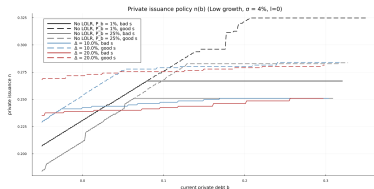
Fixed LOLR borrowing



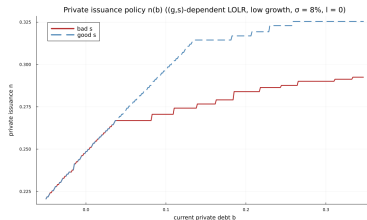
State-dependent LOLR borrowing

- Conditional private schedule at $I' = 0$
- Low growth; vary b' and plot $(n(b', 0, g, s), R(b', 0, g, s))$
- Solid: bad s Dashed: good s
- Left colors compare fixed- Δ models
- Right compares bad/good s within one state-dependent- Δ model

Private Borrowing Policy: Baseline vs V2



Fixed- Δ models



State-dependent Δ

Plot

$$n(b) = n(b^*(b), l^*(b), g, s)$$

with current $l = 0$ and low growth.

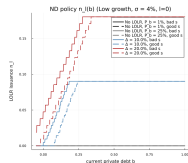
Left: baseline

- Separate fixed- Δ models; colors index Δ .
- Solid/dashed split bad and good s .

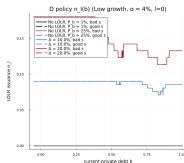
Right: V2

- One model with state-dependent LOLR cap.
- Difference between solid and dashed now comes directly from changing the available LOLR cap across sunspots.

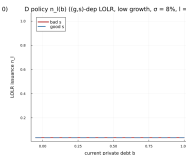
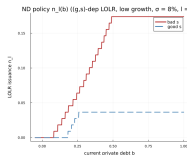
LOLR Borrowing Policy: Baseline vs V2



Fixed-Δ models



State-dependent Δ



- Left subpanel:

$$n_l^{ND}(b) = \frac{g^{I_{ND}^*}(b)}{R_l^{ND}}$$

- Right subpanel:

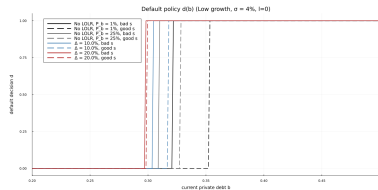
$$n_l^D(b) = \frac{g^{I_D^*}(b)}{R_l^D}$$

- Both use the full current b grid with current $l = 0$.

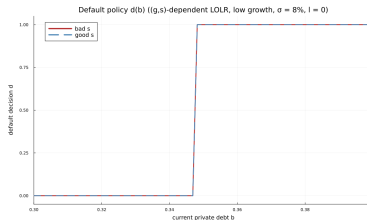
Difference

- Baseline colors compare fixed-Δ models.
- V2 compares bad and good s within one state-dependent-Δ model, so the sunspot itself changes LOLR access in

Default Regions: Baseline vs V2



Fixed- Δ models



State-dependent Δ

Plot

$$d(b, 0, g, s, \epsilon) = \mathbf{1}\{V^D > V^{ND}\}$$

over the full current b grid at low growth.

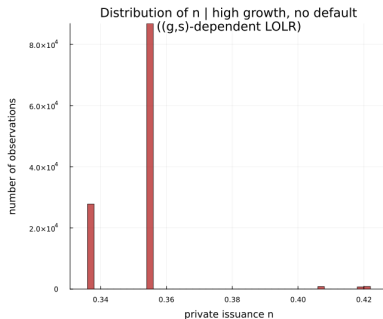
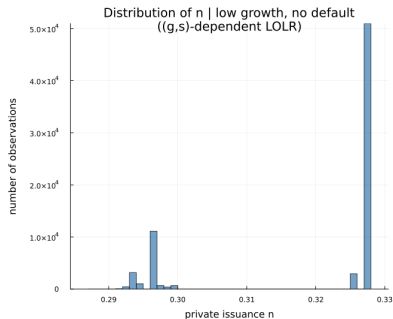
Left: baseline

- Different colors are different constant- Δ models.

Right: V2

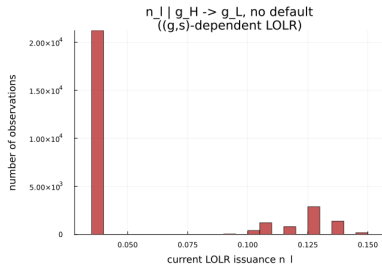
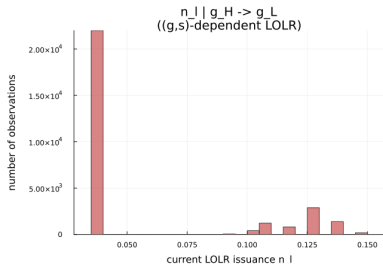
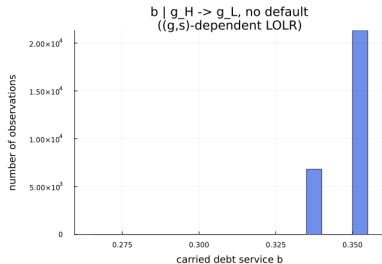
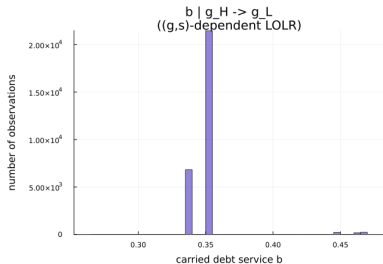
- Same model in both lines.
- Bad s has higher repayment LOLR cap than good s , so any shift is the direct within-model effect of state-dependent access.

State-Dependent LOLR: Distribution of Private Issuance



- Histogram of simulated n_t
- Low-growth panel on the left
- High-growth panel on the right
- No-default observations only
- Bars show number of observations, not density

State-Dependent LOLR: $g_H \rightarrow g_L$ Distributions



- Top row: carried debt service b_t
- Bottom row: current LOLR issuance $n_{l,t}$
- Left column: all $g_H \rightarrow g_L$ transitions
- Right column: $g_H \rightarrow g_L$ transitions with no default
- Bars show number of observations, not density

Appendix: Simulation Variables

$$f_t = \underbrace{g_t}_{\substack{\text{scale} \\ \text{not discounting}}} \underbrace{b'_t}_{\substack{\text{principal payment} \\ \text{based on budget constraint}}}$$

$$q_t = \frac{1}{R_t} = \frac{E_t[Q_{t+1}]}{1 + r^*}$$

$$q_t b_t = \frac{b_t}{R_t}$$

$$n_t = q_t f_t = q_t g_t b'_t = \frac{g_t b'_t}{R_t}$$

- f_t is the face value of newly issued private debt.
- n_t is private proceeds today from newly issued debt.
- $q_t b_t$ is the market value of inherited private debt.

Appendix: Default Transition Table

Table: Default-transition shares after burn-in (percent)

Moment	No $P_b = 1\%$	LOLR $P_b = 25\%$	$\Delta = 1\%,$ $\sigma = 4\%$	$P_b = 25\%$ $\sigma = 8\%$	$\Delta = 5\%,$ $\sigma = 4\%$	$P_b = 25\%$ $\sigma = 8\%$	$\Delta = 10\%,$ $\sigma = 4\%$	$P_b = 25\%$ $\sigma = 8\%$	$\Delta = 20\%,$ $\sigma = 4\%$	$P_b = 25\%$ $\sigma = 8\%$	Main $\sigma = 8\%, \Delta_{ND, LB} = 20\%, \Delta_{LB} = 10\%$
% of default starts by $g_H \rightarrow g_L$	61.11	99.93	99.83	99.86	99.69	99.89	99.86	99.83	99.61	99.72	77.74
% of $g_H \rightarrow g_L$ ends in default	1.90	99.79	100.00	99.89	99.79	99.86	99.96	99.79	99.93	99.79	2.69

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Appendix: All Moments Table

Moment	Main, $\sigma = 8\%$ $\Delta_{ND,LB} = 20\%, \Delta_{else} = 5\%$	S8-15/0, $\sigma = 8\%$ $\Delta_{ND,LB} = 15\%, \Delta_{else} = 0\%$	S8-10/0, $\sigma = 8\%$ $\Delta_{ND,LB} = 10\%, \Delta_{else} = 0\%$	S4-20/5, $\sigma = 4\%$ $\Delta_{ND,LB} = 20\%, \Delta_{else} = 5\%$	S4-15/0, $\sigma = 4\%$ $\Delta_{ND,LB} = 15\%, \Delta_{else} = 0\%$	S4-30/0, $\sigma = 4\%$ $\Delta_{ND,LB} = 30\%, \Delta_{else} = 0\%$
<i>First moments (percent)</i>						
avg(spread)	0.0021	0.0051	0.0030	0.0009	0.0012	0.0012
avg(qb/y)	0.3334	0.3330	0.3324	0.3372	0.3371	0.3374
avg(f/y)	0.3492	0.3512	0.3489	0.3524	0.3524	0.3527
avg(n/y)	0.3366	0.3373	0.3360	0.3401	0.3400	0.3403
avg(n.l/y)	0.0473	0.0090	0.0081	0.0496	0.0102	0.0104
avg(l/y)	0.0524	0.0101	0.0091	0.0530	0.0110	0.0113
avg(b/y)	0.3457	0.3465	0.3451	0.3493	0.3493	0.3495
avg(tb/y)	0.0143	0.0103	0.0102	0.0127	0.0101	0.0101
default rate	0.0053	0.0117	0.0072	0.0021	0.0026	0.0028
<i>Low-growth state (percent)</i>						
avg(spread)	0.0011	0.0001	0.0000	0.0000	0.0000	0.0000
avg(qb/y)	0.3483	0.3466	0.3467	0.3527	0.3524	0.3527
avg(f/y)	0.3450	0.3427	0.3426	0.3486	0.3483	0.3486
avg(n/y)	0.3330	0.3311	0.3310	0.3369	0.3366	0.3368
avg(n.l/y)	0.0622	0.0240	0.0212	0.0657	0.0265	0.0272
avg(l/y)	0.0610	0.0168	0.0149	0.0618	0.0178	0.0182
avg(b/y)	0.3609	0.3588	0.3589	0.3651	0.3647	0.3650
avg(tb/y)	0.0267	0.0203	0.0215	0.0243	0.0194	0.0193
default rate	0.0138	0.0304	0.0187	0.0055	0.0069	0.0071
<i>High-growth state (percent)</i>						
avg(spread)	0.0028	0.0082	0.0049	0.0015	0.0020	0.0020
avg(qb/y)	0.3242	0.3249	0.3236	0.3276	0.3275	0.3278
avg(f/y)	0.3518	0.3563	0.3528	0.3547	0.3549	0.3552
avg(n/y)	0.3388	0.3410	0.3390	0.3421	0.3421	0.3424
avg(n.l/y)	0.0381	0.0000	0.0000	0.0395	0.0000	0.0000
avg(l/y)	0.0472	0.0061	0.0055	0.0476	0.0068	0.0070
avg(b/y)	0.3365	0.3391	0.3367	0.3395	0.3396	0.3399
avg(tb/y)	0.0067	0.0042	0.0032	0.0054	0.0043	0.0045
default rate	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

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